

TEST CASES OF DECISION TABLES AND STATE TRANSITIONS

Name: Divay Garg (Team7)

Employee code: 85010911

Part 1: Decision Table Based Test Cases

1. Login System Decision Table

Decision Table Matrix (Image)

ID	CONDITIONS/ACTIONS	TEST CASE 1	TEST CASE 2	TEST CASE 3	TEST CASE 4
Condition 1	Valid User ID	T	T	F	F
Condition 2	Valid Password	T	F	T	F
Action 1	Home Page	Execute			
Action 2	Show a Message as 'Invalid User Credentials'		Execute	Execute	Execute

Source: *SoftwareTestingMaterial – Decision Table Test Case Design Technique (Login Example)* - <https://www.softwaretestingmaterial.com/decision-table-test-design-technique/>

Decision Table Conditions

C1: Username Valid

C2: Password Valid

Test Cases Derived from the Table

TC ID	Test Steps	Test Data	Expected Result	Rule Covered
TC1_01	Enter valid username & password, click Login	Username: valid, Password: valid	Access granted, redirected to dashboard	R1
TC1_02	Enter valid username, invalid password	Username: valid, Password: invalid	Error: "Incorrect password"	R2
TC1_03	Enter invalid username, valid password	Username: invalid, Password: valid	Error: "Invalid username"	R3
TC1_04	Enter invalid username and password	Both invalid	Error: "Invalid username" (shown first)	R4

Individual Explanation of Each Test Case

TC1_01: Validates the happy path where both credentials are correct. Access should be granted.

TC1_02: Confirms a correct username with a wrong password shows a password-specific error.

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TC1_03: Confirms an incorrect username is detected even with a correct password.

TC1_04: Verifies that when both fields are wrong, username validation takes priority in the error shown.

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2. Triangle Classification Decision Table

Decision Table Matrix (Image)

	1	2	3	4	5	6	7	8	9	10	11
C1-1: $a < b+c$?	F	T	T	T	T	T	T	T	T	T	T
C1-2: $b < a+c$?	-	F	T	T	T	T	T	T	T	T	T
C1-3: $c < a+b$?	-	-	F	T	T	T	T	T	T	T	T
C2: $a = b$?	-	-	-	T	T	T	T	F	F	F	F
C3: $a = c$?	-	-	-	T	T	F	F	T	T	F	F
C4: $b = c$?	-	-	-	T	F	T	F	T	F	T	F
A1: Not a Triangle	X	X	X								
A2: Scalene											X
A3: Isosceles							X		X	X	
A4: Equilateral				X							
A5: Impossible					X	X		X			

Decision Table Conditions

C1: All Three Sides Equal

C2: Exactly Two Sides Equal

C3: Sides Satisfy Triangle Inequality (valid triangle)

Test Cases Derived from the Table

TC ID	Test Steps	Test Data	Expected Result	Rule Covered
TC2_01	Enter three equal sides	A=5, B=5, C=5	"Equilateral Triangle"	R1
TC2_02	Enter two equal sides	A=5, B=5, C=8	"Isosceles Triangle"	R2
TC2_03	Enter three different, valid sides	A=3, B=4, C=5	"Scalene Triangle"	R3
TC2_04	Enter sides that cannot form a triangle	A=1, B=1, C=10	"Not a Valid Triangle"	R4

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Individual Explanation of Each Test Case

TC2_01: Validates classification when all three sides are equal.

TC2_02: Validates classification when exactly two sides are equal.

TC2_03: Validates classification when all sides differ but still form a valid triangle.

TC2_04: Ensures sides violating the triangle inequality rule are rejected as invalid, regardless of equality pattern.

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3. Single-Trip Insurance Premium Decision Table

		Rules							
Conditions		1	2	3	4	5	6	7	8
Condition	→ C1. Destination	EU	EU	EU	EU	Non-EU	Non-EU	Non-EU	Non-EU
	C2. Period	< 1w	< 1w	>=1W	>=1W	< 1w	< 1w	>=1W	>=1W
	C3. Adult?	Y	N	Y	N	Y	N	Y	N
Actions		1	2	3	4	5	6	7	8
Action	→ A1. Premium (£)	46	53	60	67	54	61	68	75

This is the premium when all the conditions stated under this rule are met.

Decision Table Conditions

C1: Age ≤ 60 Years

C2: Trip Duration ≤ 14 Days

C3: Coverage Type = Comprehensive

Test Cases Derived from the Table

TC ID	Test Steps	Test Data	Expected Result	Rule Covered
TC3_01	Select age 25, 7-day trip, Basic coverage	Age=25, Days=7, Cover=Basic	Standard premium applied	R1
TC3_02	Select age 25, 7-day trip, Comprehensive coverage	Age=25, Days=7, Cover=Comprehensive	Standard premium + comprehensive loading	R2
TC3_03	Select age 65 (senior), 7-day trip	Age=65, Days=7	Senior age loading applied	R3
TC3_04	Select age 25, 30-day trip	Age=25, Days=30	Long-duration trip loading applied	R4
TC3_05	Select age 70, 30-day trip, Comprehensive	Age=70, Days=30, Cover=Comprehensive	Maximum premium tier applied	R5

Individual Explanation of Each Test Case

TC3_01: Validates the base case — young traveler, short trip, basic cover — with standard

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premium.

TC3_02: Confirms comprehensive coverage adds a loading on top of the standard premium.

TC3_03: Verifies senior travelers (>60) are charged an age-based loading.

TC3_04: Verifies trips longer than 14 days incur a duration-based loading.

TC3_05: Confirms that stacking all loading factors (senior + long trip + comprehensive) results in the maximum premium tier.

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4. Loan Approval Decision Table

		Rules					
Conditions		1	2	3	4	5	6
Condition	→ C1. Address proof provided	N		Y	Y	Y	Y
	C2. Identity proof provided		N	Y	Y	Y	Y
	C3. Loan amount < monthly salary			Y			
	C4. Loan amount ≥ monthly salary					Y	Y
	C5. Loan purpose				Home purchase	Pay tax	Other
	C6. Home owner?				Y		
Actions		1	2	3	4	5	6
Action	→ A1. Approve loan request immediately			X	X	X	
	A2. Review loan request manually						X
	A3. Reject loan request	X	X				

Rule

Condition

Action

Action(s) marked "X" will be taken when all the conditions marked "Y" are met.

Decision Table Conditions

C1: Credit Score ≥ 700

C2: Income Sufficient

C3: Existing Debt Low

Test Cases Derived from the Table

TC ID	Test Steps	Test Data	Expected Result	Rule Covered
TC4_01	High credit score, sufficient income, low debt	Score=800, Income=High, Debt=Low	"Loan Approved"	R1
TC4_02	Low credit score, high income, low debt	Score=550, Income=High, Debt=Low	"Loan Rejected"	R2
TC4_03	High credit score, insufficient income	Score=800, Income=Low	"Loan Rejected"	R3
TC4_04	High score, sufficient income, high debt	Score=750, Income=High, Debt=High	"Referred for Manual Review"	R4

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Individual Explanation of Each Test Case

TC4_01: Validates the happy path where all three conditions favour approval.

TC4_02: Confirms a low credit score rejects the loan even with good income and low debt.

TC4_03: Confirms insufficient income rejects the loan despite a strong credit score.

TC4_04: Verifies that high existing debt routes an otherwise-qualifying application to manual review rather than auto-approval.

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5. E-Commerce Discount Eligibility Decision Table

Discount Table	Type of customer	Customer paid cash	Number of units	Discount
Rule 1	Wholesale	Y	<50	4
Rule 2	Wholesale	Y	>=50	6
Rule 3	Wholesale	N	<50	2
Rule 4	Wholesale	N	>=50	4
Rule 5	Retail	Y	<50	2
Rule 6	Retail	Y	>=50	4
Rule 7	Retail	N	<50	0
Rule 8	Retail	N	>=50	2

Decision Table Conditions

C1: Cart Value $\geq$ ₹1000

C2: Coupon Applied

C3: Member

Test Cases Derived from the Table

TC ID	Test Steps	Test Data	Expected Result	Rule Covered
TC5_01	Cart $\geq$ ₹1000, coupon applied, member	Cart=₹1500, Coupon=Yes, Member=Yes	Maximum discount applied	R1
TC5_02	Cart $\geq$ ₹1000, no coupon, non-member	Cart=₹1500, Coupon=No, Member=No	No discount applied	R2
TC5_03	Cart < ₹1000, coupon applied	Cart=₹500, Coupon=Yes	Coupon rejected: "Minimum cart value not met"	R3
TC5_04	Cart $\geq$ ₹1000, no coupon, member	Cart=₹1500, Coupon=No, Member=Yes	Member-only discount applied	R4

Individual Explanation of Each Test Case

TC5_01: Validates the best-case scenario where cart value, coupon, and membership all qualify for maximum savings.

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TC5_02: Confirms no discount is given when none of the qualifying conditions are met.

TC5_03: Ensures a coupon is rejected outright when the minimum cart value isn't reached, regardless of membership.

TC5_04: Verifies members receive a discount automatically, even without applying a coupon.

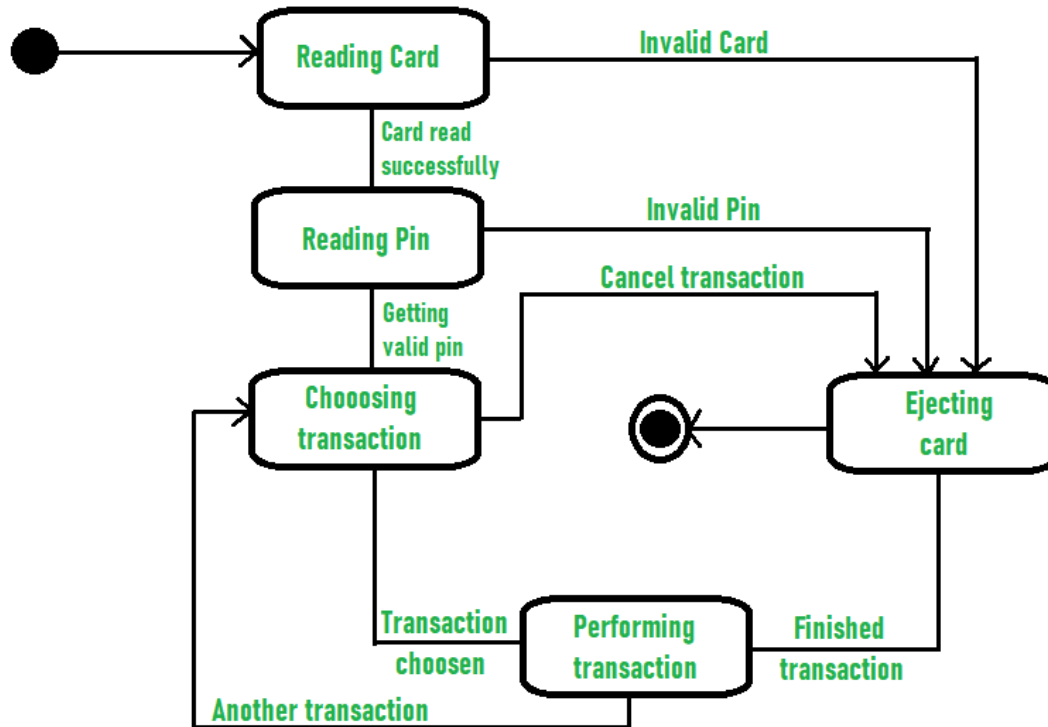
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Part 2: State Transition Diagram Based Test Cases

1. ATM System State Transition



State Transition Diagram for ATM System

Test Cases Derived from the Diagram

TC ID	Test Steps	Test Data / Trigger	Expected Result
TC1_01	Insert card at Idle state	Card inserted	Transitions to "PIN Entry" state
TC1_02	Enter correct PIN in "PIN Entry" state	PIN = correct	Transitions to "Transaction" state
TC1_03	Enter incorrect PIN 3 times	PIN = wrong (x3)	Transitions to "Card Retained/Blocked" state
TC1_04	Complete withdrawal, then remove card	Transaction complete	Transitions back to "Idle" state
TC1_05	Try entering PIN without inserting card	No card inserted	Invalid transition — system stays in "Idle", no PIN screen shown

Explanation

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TC1_01 , TC_02, TC1_04,: Test every valid transition in the ATM's normal cycle (Idle → PIN Entry → Transaction → Idle).

TC1_03: Tests the boundary/edge transition where repeated PIN failures move the system to a blocked state instead of remaining in "PIN Entry".

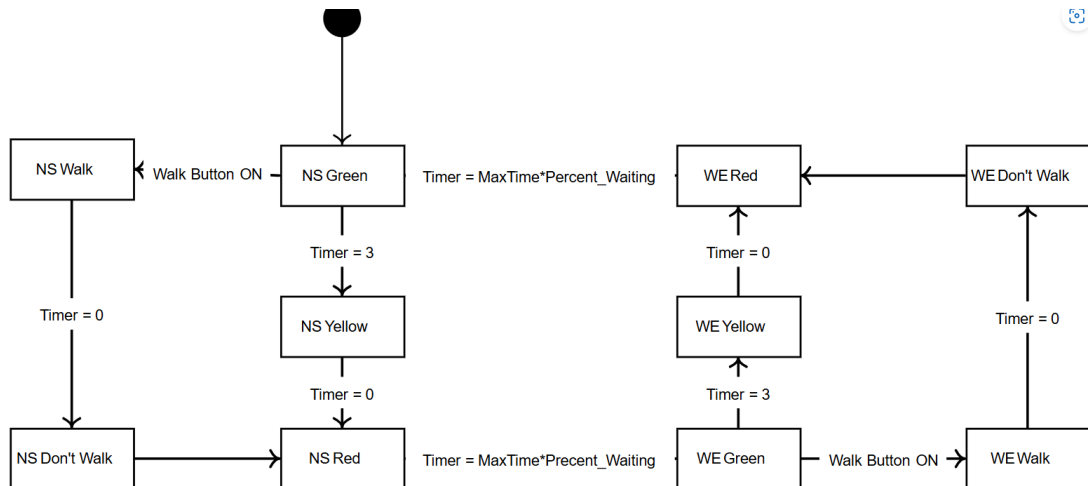
TC1_05: Tests an invalid transition to confirm the system cannot skip states out of sequence (no PIN entry without a card).

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2. Traffic Light System State Transition



Test Cases Derived from the Diagram

TC ID	Test Steps	Test Data / Trigger	Expected Result
TC2_01	Observe light at Red, wait for timer	Timer expires	Transitions to "Green"
TC2_02	Observe light at Green, wait for timer	Timer expires	Transitions to "Yellow"
TC2_03	Observe light at Yellow, wait for timer	Timer expires	Transitions to "Red"
TC2_04	Attempt to skip from Red directly to Yellow	Invalid trigger	System rejects/ignores invalid transition; stays on defined cycle

Explanation

TC2_01 to TC2_03: Validate the fixed cyclical sequence (Red → Green → Yellow → Red) that the system must always follow.

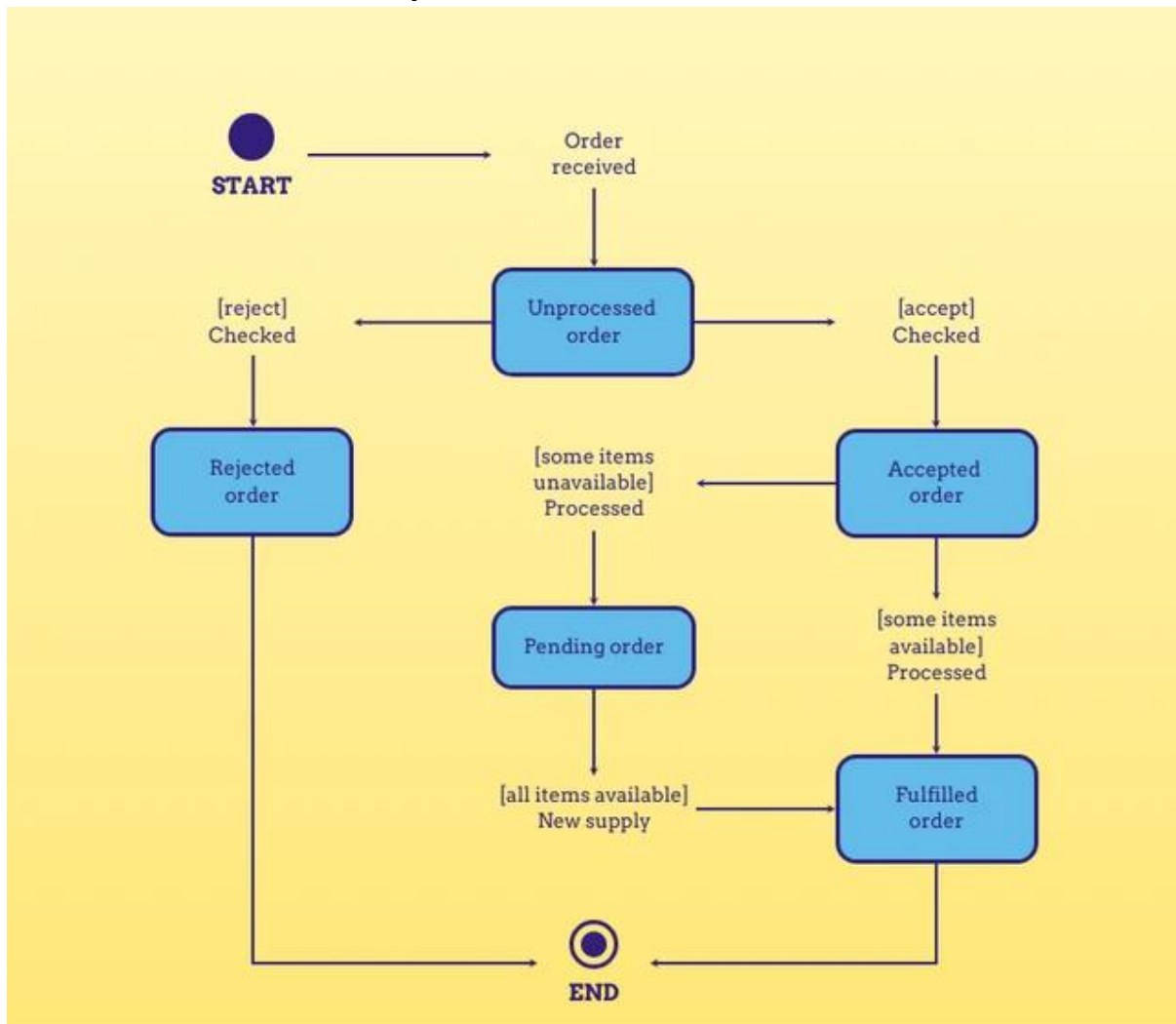
TC2_04: Confirms the system doesn't allow skipping a state out of order, since traffic lights must follow a strict one-directional cycle.

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3. E-Commerce Order Lifecycle State Transition



Test Cases Derived from the Diagram

TC ID	Test Steps	Test Data / Trigger	Expected Result
TC3_01	Place an order	Order created	State = "Placed"
TC3_02	Ship the placed order	Seller marks as shipped	State transitions "Placed" → "Shipped"
TC3_03	Mark shipped order as delivered	Delivery confirmed	State transitions "Shipped" → "Delivered"
TC3_04	Cancel an order while still "Placed"	Cancel request	State transitions "Placed" → "Cancelled"
TC3_05	Attempt to cancel an order already "Shipped"	Cancel request	Invalid transition — system blocks cancellation, shows "Cannot cancel shipped order"

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TC3_06	Attempt to cancel a "Delivered" order	Cancel request	Invalid transition — blocked, return process suggested instead
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Explanation

TC3_01 to TC3_03: Cover the full forward path of a successful order (Placed → Shipped → Delivered).

TC3_04: Covers the valid early-exit path where an order is cancelled before shipping.

TC3_05 and TC3_06: Test invalid transitions to confirm the system correctly restricts cancellation once an order has moved past the "Placed" state.

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4. Login/Logout System State Transition

User System Login State Diagram



Test Cases Derived from the Diagram

TC ID	Test Steps	Test Data / Trigger	Expected Result
TC4_01	Submit valid credentials from "Logged Out"	Credentials = correct	Transitions "Logged Out" → "Logged In"
TC4_02	Submit invalid credentials	Credentials = incorrect	Stays in "Logged Out" state, error shown
TC4_03	Click Logout while "Logged In"	Logout triggered	Transitions "Logged In" → "Logged Out"

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TC4_04	Session times out while "Logged In"	Inactivity timer expires	Auto-transitions "Logged In" → "Logged Out"
TC4_05	Try accessing account page while in "Logged Out" state	Direct URL/page access attempt	System blocks access, redirects to "Logged Out" home/login prompt

Explanation

TC4_01 and TC4_03: Validate the two primary transitions — successful login and manual logout.

TC4_02: Confirms invalid credentials keep the system in "Logged Out" rather than incorrectly transitioning.

TC4_04: Tests the automatic, timeout-driven transition unique to session-based systems.

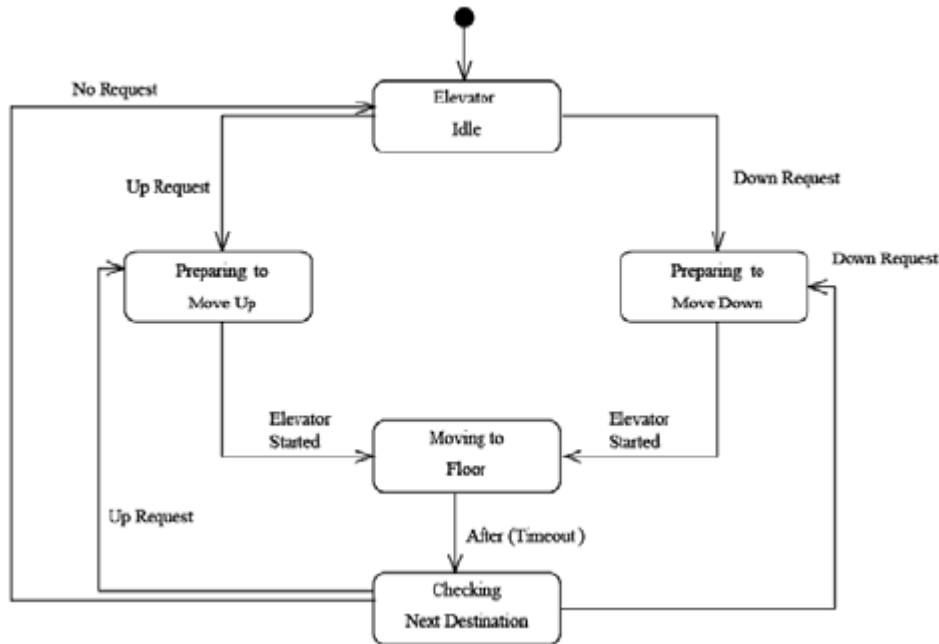
TC4_05: Confirms a "Logged Out" state cannot reach pages meant only for "Logged In" users, enforcing state-based access control.

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5. Elevator Controller State Transition



Test Cases Derived from the Diagram

TC ID	Test Steps	Test Data / Trigger	Expected Result
TC5_01	Press floor button while Idle at ground floor	Requested floor > current floor	Transitions "Idle" → "Moving Up"
TC5_02	Elevator reaches requested floor while "Moving Up"	Floor reached	Transitions "Moving Up" → "Door Open"
TC5_03	Door remains open, timer expires	Timer expires, no new request	Transitions "Door Open" → "Idle"
TC5_04	Press floor button below current floor	Requested floor < current floor	Transitions "Idle" → "Moving Down"
TC5_05	Press door-open button while elevator is "Moving Up"	Invalid trigger mid-transit	System ignores/rejects command; state remains "Moving Up" until floor reached

Explanation

TC5_01 and TC5_04: Cover both directional movement paths (up and down) from the Idle state.

TC5_02 and TC5_03: Validate the return-to-idle cycle once a request is fulfilled and the door timer lapses.

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TC5_05: Tests a safety-critical invalid transition — the door should not open while the elevator is between floors.